Spectral integrity

PARTIAL: grids/masks valid; absolute scales missing

Material coverage

PARTIAL: Rock/Bare/dead labels; water/snow/litter/shade gaps

Representativeness

PARTIAL: library holdouts, no same-site full imagery

Distinguishability

CONDITIONAL: sampled-group differences previously shown

Mixture identifiability

TESTED: substantial other-class convex ambiguity

Fraction recovery

TESTED: conditional synthetic fraction errors, no real truth

Transferability

TESTED: bounded grouped holdouts; some unavailable

Sensor robustness

TESTED: paired synthetic OLI only; no abundance ground truth

Uncertainty calibration

NOT ESTABLISHED: empirical bundles not calibrated intervals

Computation

FEASIBLE: bounded candidates, exact search and checkpoints

Operational material-abundance readiness

Green vegetation	INSUFFICIENT EVIDENCE
Deciduous	INSUFFICIENT EVIDENCE
Evergreen	INSUFFICIENT EVIDENCE
Herbaceous	INSUFFICIENT EVIDENCE
Woody shrub	INSUFFICIENT EVIDENCE
NPV / dead	INSUFFICIENT EVIDENCE
Rock	INSUFFICIENT EVIDENCE
Bare (not verified soil)	INSUFFICIENT EVIDENCE
Soil	INSUFFICIENT EVIDENCE
Water / snow	INSUFFICIENT EVIDENCE
Measured shade	INSUFFICIENT EVIDENCE
Species-level	INSUFFICIENT EVIDENCE

Same status for NEON and Landsat-like abundance.
No cell is promoted to READY from a synthetic-only fit.

No opaque readiness score. Current support is for labelled-library diagnostics and explicitly conditional synthetic mixtures, not operational ecological abundance.
Missing source-scale/purity/reference fractions are blocking data problems. Each material conclusion is tied to quantitative tables or explicitly insufficient evidence.